LESSON PLAN

AIRCRAFT PROPULSION & ENGINEERING SYSTEM (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	14	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Restoring surface conversion coatings, procedures and precautions.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Restoring surface conversion coatings, procedures and precautions.
5. Specific Objectives	6.1.3.1 Trainee will be able to perform rectification of corrosion.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of surface preparation and coating application on test panels; Safety precautions regarding PPE and ventilation; Instructor supervision of trainee application technique.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary causes of corrosion on aircraft surfaces?
(b) Why is surface preparation critical before applying conversion coatings?
(c) List three safety precautions required when handling chemical coatings.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Inspection and assessment of corroded surfaces • Mechanical removal of loose corrosion products • Chemical cleaning and neutralization procedures • Surface drying and contamination checks • Application of primer and conversion coating	L & Ds	PPT, IB	15 min
(b)	• Practical application of rectification procedures • Verification of coating thickness and adhesion • Identification of common application defects • Corrective actions for pinholes and runs • Final inspection and quality sign-off	L & Ds	PPT, PDF	20 min
(c)	• Review of environmental constraints and storage • Documentation of rectification work performed • Precautions for handling hazardous waste • Transition to engine component corrosion treatment • Summary of key procedural steps	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between mechanical and chemical removal of corrosion.
(b) Describe the correct sequence for applying a conversion coating to ensure adhesion.

12. Home Assignment and Follow Up	Prepare a brief report outlining the specific precautions required when restoring surface conversion coatings on engine components, referencing the procedures covered in this lesson and the upcoming lesson on engine component corrosion treatment.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Corrosion treatment of engine components.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min